YIFEI SUN

Principal Software Engineer · Backend Platform, Payments & Distributed Systems · 13 Years

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SUMMARY

Principal-level backend & platform engineer with 13 years building fault-tolerant distributed systems and high-traffic platform infrastructure. Led the redesign of a critical API orchestration layer serving millions of daily requests, reducing p99 latency 5× and achieving 99.9% availability across 10+ teams. Built correctness verification frameworks, event-driven architectures (Kafka, CQRS), and observability standards adopted org-wide. Track record of building platform infrastructure that multiple teams depend on, driving cross-functional alignment across departments, and delivering complex 0→1 systems from inception through production.

LEADERSHIP HIGHLIGHTS

- Led cross-org platform modernization spanning 10+ engineering teams across 3 departments with measurable reliability impact
- Designed event-driven architectures (Kafka, CQRS) and observability standards (distributed tracing, SLO-based latency budgets)
- Built correctness verification framework — automated validation at 50% production traffic for safe, data-driven migrations
- Mentored senior engineers toward Principal-level scope; drove architectural standards via RFCs and design docs

Technologies: Java, Spring Boot, Python, Node.js · Service Orchestration, API Design, Fault Tolerance, Idempotency · AWS, Kubernetes, Docker, Kafka, Redis, PostgreSQL · GraphQL (Netflix DGS), Microservices, Event-Driven, CQRS · Datadog, Grafana, Distributed Tracing · Keycloak, SSO, OAuth 2.0

INDEPENDENT PROJECT

GenAI Decision Engine & Experimentation Platform

- Architected a hybrid rules + multi-path GenAI engine (fast vs. deep inference) with a policy overlay over heterogeneous telemetry, producing explainable real-time recommendations.
- Built "Prompt Lab" for end-to-end decision simulation, offline evaluation/backtesting, and prompt/parameter search (grid & Bayesian optimization) with multi-provider LLM support.
- Designed a configurable policy framework with DB-backed configs, environment-specific overrides, and a scheduler-driven tuning loop that sweeps parameters and publishes promotable config versions.

EXPERIENCE

CarGurus — Principal Software Engineer / Technical Lead

Oct 2019 – Present

High-Traffic Platform Modernization — Orchestration & Fault Tolerance

- Led 3 principal/senior engineers and partnered with ~10 teams across 3 departments to redesign the core platform behind CarGurus' most-visited page (~8,000 rpm, millions of daily impressions).
- Introduced a new orchestration service consolidating 12 downstream integrations into 5 well-defined services with circuit breaking, fault isolation, and idempotent retry — defining the platform infrastructure that teams across the org depend on.
- Produced initial architecture in ~2 weeks; shipped first milestone in 6 months; completed ahead of schedule.

p99: 250 ms → 50 ms (5×) **Availability: 98% → 99.9%** **Deploy: 1 hr → 3 min** (20×)

GraphQL Federation — Unified Serving Layer

- Designed and integrated a Netflix DGS subgraph into CarGurus' federated GraphQL gateway, creating a unified API layer for frontend, backend, and AI agent consumers from a single graph.
- Implemented schema federation, batch data loaders (N+1 prevention), and JVM warmup strategies; built load tests and staging integration tests for production-grade reliability.

Digital Commerce Platform — 0→1 Under Ambiguity

- Led backend design and launch of multiple greenfield products using microservice and event-driven architecture; enabled the company to rapidly test, launch, and safely retire product experiments without destabilizing core systems.

Parity Testing Framework — Correctness Verification at Scale

- Built an automated correctness verification framework that validated new system output against legacy at 50% production traffic — catching data integrity issues before they reached users and enabling safe, data-driven migration rollout.

Cross-Org Platform Leadership & Technical Direction

- Defined architectural standards — API design, caching, distributed orchestration, observability — adopted across multiple teams via RFCs and design docs.
- Mentored senior engineers on system design, performance, and path to Principal-level scope.

AI Integration — MCP Server & GenAI Platform

- Sole backend engineer for an MCP server on the OpenAI Apps SDK, exposing platform capabilities as tools callable by AI agents. Separately built a GenAI automation platform with multi-provider LLM orchestration, RAG-augmented inference, and an evaluation harness.

Applause — Senior Software Engineer / Technical Lead

Oct 2017 – Oct 2019

Authentication & Identity Platform

- Upgraded legacy authentication to a modern Keycloak-based SSO platform, unifying auth across internal systems, an enterprise partner platform, and a large external community. Migrated multiple Angular, Node.js, and Java monoliths behind the new auth gateway with minimal downtime.

Event Sourcing & CQRS Platform Initiative

- Drove design and prototyping of a microservice architecture based on event sourcing and CQRS; defined foundational patterns for services, events, and read/write models adopted by new domains.

Cross-Org Collaboration — Customer Support & Workflow Automation

- Partnered with product, support, and customer-facing teams to modernize internal workflows and enterprise integrations with clear observability and rollback paths.

Toast Inc. — Senior Software Engineer

Sep 2014 – Oct 2017

Monolith → Microservices Decomposition

- Among the earliest engineers during hypergrowth; designed and implemented core backend services (Order Service, Labor Service, Payment Integration APIs) processing restaurant transactions end-to-end, becoming the platform other teams built upon.
- Built initial AWS deployment pipelines and automated testing framework, enabling safer releases during rapid product expansion.

Third-Party Integrations & Cross-Functional Leadership

- Led integrations with key payment and workforce partners, ensuring reliable bidirectional sync, schema evolution, and transaction reconciliation. Partnered with PMs, sales, and support to translate customer feedback into platform capabilities.

The MathWorks Inc. — Software Engineer II

Jul 2013 – Aug 2014

Developed internal engineering tools for MATLAB/Simulink workflows. Improved backend automation systems for build/test reliability and developer productivity.

EDUCATION

M.S. in Electrical and Computer Engineering — Northeastern University, Boston 2010 – 2013

B.S. in Information Science — Beijing University of Aeronautics and Astronautics 2006 – 2010